# Band-1 Hadamard Order 9240 – Classical Paley-I (q=9239); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9240 Paley-I PASS; Paley-II SKIP (q=4619=31*149 NOT prime power); Kronecker FAIL/SKIP (4*2310; 2310===2 mod 4 NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

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**Document class.** Scientific computational / construction certificate (Band-1).
**Program.** Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
**Parents.** `2026-09-21_Band1_Hadamard_9204_PaleyI`; `2026-09-21_Band1_Hadamard_9200_PaleyI`; `2026-09-21_Band1_Hadamard_9188_PaleyI`.
**Lab.** `/workspace/experiments/band1/hadamard_9240_paleyI/`.
**Cutoff.** 2026-09-21T23:45:32Z (~7:45PM ET).

**Honesty stamp (non-negotiable).**
This memo claims **finite construction of order 9240 only (verified max|HHˆT-nI|=0), via classical Paley-I q=9239 (prime ===3 mod 4). 9240 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9240: Paley-I **PASS**s ($q = 9239$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL** ($q = 4619 = 7 \cdot 659$ NOT prime power); Kronecker **FAIL/SKIP** (note $9240 = 4 \cdot 2310$; $2310 \equiv 3 \pmod 4$ NOT Hadamard-admissible). **9240** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9204+9200+9188+9152+9104+9128+9124+9092+2220+8884+2236+2160+2140+8564 / 1064+2128 / ...** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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## Abstract

Finite construction of order **9240**.

- $H_{9240}$ = Paley-I ($q = 9239$) – classical SPK
- Paley-II for 9240: **SKIP** ($q = 4619 = 7 \cdot 659$ NOT prime power)
- Kronecker for 9240: **FAIL/SKIP** ($4 \cdot 2310$; $2310 \equiv 3 \pmod 4$ NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

**SPK: IN SPK via Paley-I**.
**Gap 2052:** OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_9240.py`

**Results.** overall_pass=True; units 114/114; wall 642.4144s; constructed max|HHˆT-nI|=0.

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## Arithmetic

- Target $n = 9240 = q + 1$ with $q = 9239$ prime $\equiv 3 \pmod 4$ -> Paley-I eligible.
- Paley-II: $q = 4619 = 7 \cdot 659$ NOT prime power -> SKIP.
- Kronecker: $9240 = 4 \cdot 2310$; $2310 \equiv 3 \pmod 4$ NOT Hadamard-admissible – FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

## Constructions (verified)

| Object | Method | max\|HHˆT-nI\| | Wall (s) |
|---|---|---|---|
| H_9240 Paley-I | Paley-I q=9239 | 0 | 634.5827 |
| H_9240 Paley-II | SKIP q=4619=31*149 NOT prime power | – | – |
| H_9240 Kronecker | FAIL/SKIP (4*2310; 2310===2 mod 4 NOT Hadamard-admissible) | – | – |

## SPK closure note

Seeds include prior SPK through **9204** lineage (**9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; **1064+2128**; . . . ; **NOT** 8948/8644/8516/. . . ) + verified **9240**. Next constructive gap **2052** (OPEN).

| Quantity | Value |
|---|---|
| 9240 in SPK | **True** (Paley-I q=9239) |
| Gap 2052 | **OPEN** |
| clay_closed | false |

| Quantity | Value |
|---|---|
| GATE-04 | LOCKED |

## Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

## Next optional

q=9283->n=9284 Paley-I DIRECT q=9283 OR q=9311->n=9312 Paley-I DIRECT q=9311 OR n=2052 standard SPK constructive gap – method-attack / lit OR q=9319->n=9320 Paley-I DIRECT q=9319 OR method-attack SPK gap 2052 OR Track B / Object 1 meter await